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Center of Mass and Collision

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	40 questions
Answer key	at the end

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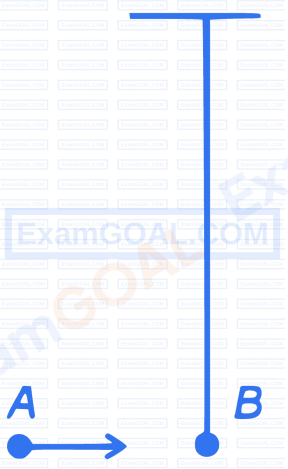
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MCQ (Single Correct Answer) 40 questions

Q1. RE-NEET 2026

Open solution

Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m , as shown in the figure. Point mass A of mass m travelling horizontally with speed $10\ \mathrm{ms}^{-1}$ hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ \mathrm{ms}^{-2}$ and neglecting the size of the bob, the value of h is:



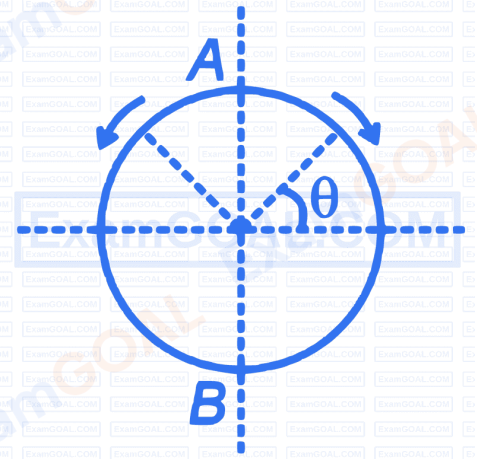
(A) 2.5 (B) 8 (C) 7 (D) 5

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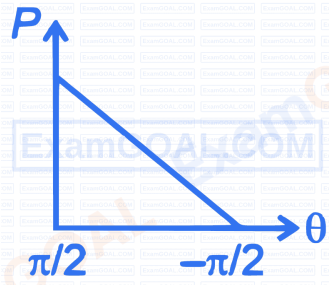
Q2. RE-NEET 2026

Open solution

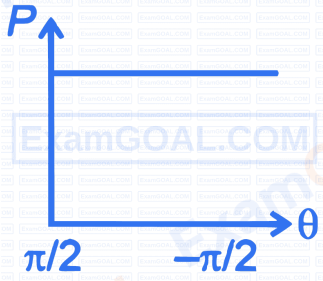
A frictionless circular wire of unit radius is fixed on the horizontal plane. Two-point particles of unit mass start moving simultaneously from point $A\left(\theta=\frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point $B\left(\theta=-\frac{\pi}{2}\right)$. During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ?



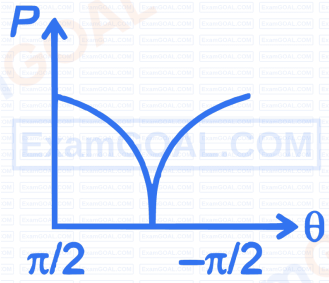
(A)



(B)

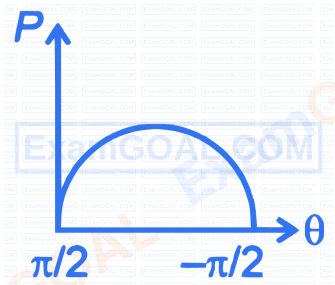


(C)



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(D)



Q3. NEET 2025

A ball of mass 0.5 kg is dropped from a height of 40 m . The ball hits the ground and rises to a height of 10 m . The impulse imparted to the ball during its collision with the ground is (Take $g = 9.8\,\mathrm{m/s^2}$)

(A) 0 (B) 84 NS (C) 21 NS (D) 7 NS

Open solution

Q4. NEET 2024

Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is

(A) $1 : 2$ (B) $2 : 1$ (C) $4 : 1$ (D) $1 : 4$

Open solution

Q5. NEET 2023 Manipur

A bullet of mass m hits a block of mass M elastically. The transfer of energy is the maximum, when :

(A) $M = m$ (B) $M = 2m$ (C) $M << m$ (D) $M >> m$

Open solution

Q6. NEET 2023 Manipur

Two particles A and B initially at rest, move towards each other under mutual force of attraction. At an instance when the speed of A is v and speed of B is 3v, the speed of centre of mass is :

(A) 2v (B) zero (C) v (D) 4v

Open solution

Q7. NEET 2023 Manipur

A $1\,\mathrm{kg}$ object strikes a wall with velocity $1\,\mathrm{m\,s^{-1}}$ at an angle of 60° with the wall and reflects at the same angle. If it remains in contact with wall for $0.1\,\mathrm{s}$, then the force exerted on the wall is :

(A) $30\sqrt{3}\,\mathrm{N}$ (B) Zero (C) $10\sqrt{3}\,\mathrm{N}$ (D) $20\sqrt{3}\,\mathrm{N}$

Open solution

Q8. NEET 2022 Phase 2

The distance covered by a body of mass 5 g having linear momentum 0.3 kg m/s in 5 s is :

(A) 0.3 m (B) 300 m (C) 30 m (D) 3 m

Open solution

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Q9. NEET 2022 Phase 2

Open solution

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : When a fire cracker (rocket) explodes in mid air, its fragments fly in such a way that they continue moving in the same path, which the fire cracker would have followed, had it not exploded.

Reason (R) : Explosion of cracker (rocket) occurs due to internal force only and no external force acts for this explosion.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) (A) is not correct but (R) is correct

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) (A) is correct but (R) is not correct

Q10. NEET 2022 Phase 1

Open solution

A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v, the speed of the third (lighter) fragment is :

(A) v **(B)** $\sqrt{2}$v **(C)** $2\sqrt{2}$v **(D)** $3\sqrt{2}$v

Q11. NEET 2022 Phase 1

Open solution

Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is :

(A) $\frac{10}{3}$ m **(B)** $\frac{20}{3}$ m **(C)** 10 m **(D)** 5 m

Q12. NEET 2021

Open solution

A ball of mass 0.15 kg is dropped from a height 10m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is (g = 10 m/s^2) nearly :

(A) 1.4 kg m/s **(B)** 0 kg m/s **(C)** 4.2 kg m/s **(D)** 2.1 kg m/s

Q13. NEET 2020 Phase 1

Open solution

Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass.
The centre of mass of the system from the 5 kg particle is nearly at a distance of :

(A) 50 cm **(B)** 67 cm **(C)** 80 cm **(D)** 33 cm

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Q14. NEET 2019

Body A of mass 4m moving with speed u collides with another body B of mass 2 m, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is :

Open solution

(A) $\frac{5}{9}$ (B) $\frac{4}{9}$ (C) $\frac{8}{9}$ (D) $\frac{1}{9}$

Q15. NEET 2018

A moving block having mass m, collides with another stationary block having mass 4m. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v, then the value of coefficient of restitution (e) will be

Open solution

(A) 0.5 (B) 0.25 (C) 0.8 (D) 0.4

Q16. NEET 2017

Which of the following statements are correct ?

Open solution

(1) Centre of mass of a body always coincides with the centre of gravity of the body.

(2) Centre of mass of a body is the point at which the total gravitational torque on the body is zero.

(3) A couple on a body produces both translational and rotational motion in a body

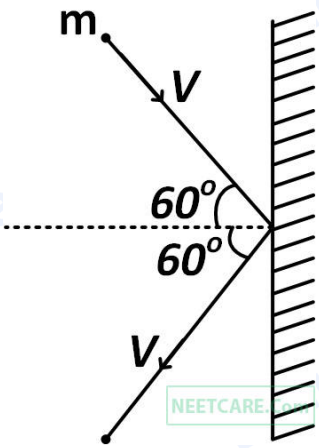
(4) Mechanical advantage greater than one means that small effort can be used to lift a large load.

(A) (1) and (2) (B) (2) and (3) (C) (3) and (4) (D) (2) and (4)

Q17. NEET 2016 Phase 2

A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure. The value of impulse imparted by the wall on the ball will be :

Open solution



(A) mV (B) $2\,mV$ (C) $\frac{mV}{2}$ (D) $\frac{mV}{3}$

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Q18. NEET 2016 Phase 2

A bullet of mass 10 g moving horizontally with a velocity of 400 m s^{-1} strikes a wood block of mass 2 kg which is suspended by light inextensible string of length 5 m. As a result, the centre of gravity of the block is found to rise a vertical distance of 10 cm. The speed of the bullet after it emerges out horizontally from the block will be

Open solution

(A) 100 m s^{-1} (B) 80 m s^{-1} (C) 120 m s^{-1} (D) 160 m s^{-1}

Q19. NEET 2016 Phase 2

Two identical balls A and B having velocities of 0.5 m s^{-1} and -0.3 m s^{-1} respectively collide elastically in one dimension. The velocities of B and A after the collision respectively will be :

Open solution

(A) -0.5 m s^{-1} and 0.3 m s^{-1} (B) 0.5 m s^{-1} and -0.3 m s^{-1}

(C) -0.3 m s^{-1} and 0.5 m s^{-1} (D) 0.3 m s^{-1} and 0.5 m s^{-1}

Q20. AIPMT 2015 Cancelled Paper

Two particles of masses m_1, m_2 move with initial velocities u_1 and u_2. On collision, one of the particles gets excited to higher level, after absorbing energy ε. If final velocities of particles be v_1 and v_2 then we must have :

Open solution

(A) $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \varepsilon = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$

(B) $\frac{1}{2}m_1^2u_1^2 + \frac{1}{2}m_2^2u_2^2 + \varepsilon = \frac{1}{2}m_1^2v_1^2 + \frac{1}{2}m_2^2v_2^2$

(C) $m_1^2u_1 + m_2^2u_2 - \varepsilon = m_1^2v_1 + m_2^2v_2$

(D) $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \varepsilon$

Q21. AIPMT 2015 Cancelled Paper

Two spherical bodies of mass M and 5M and radii R and 2R are released in free space with initial separation between their centres equal to 12R. If they attract each other due to gravitational force only, then the distance covered by the smaller body before collision is :

Open solution

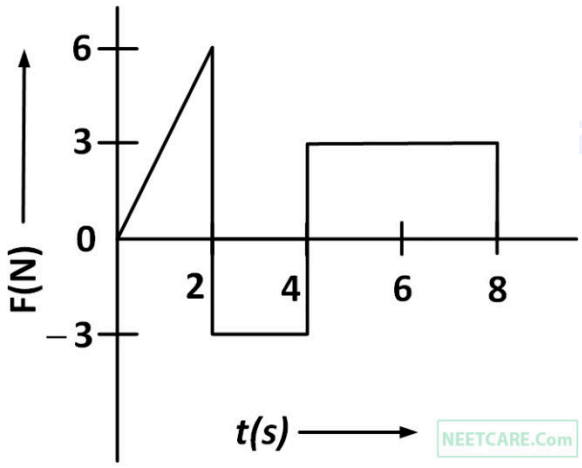
(A) 7.5R (B) 1.5R (C) 2.5R (D) 4.5R

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Q22. AIPMT 2014

The force F acting on a particle of mass m is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :

Open solution



(A) 24 N s (B) 20 N s (C) 12 N s (D) 6 N s

Q23. AIPMT 2014

A body of mass (4m) is lying in x-y plane at rest. It suddenly explodes into three pieces. Two pieces, each of mass (m) move perpendicular to each other with equal speeds (v). The total kinetic energy generated due to explosion is :

Open solution

(A) mv^2 (B) $\frac{3}{2}mv^2$ (C) $2mv^2$ (D) $4mv^2$

Q24. NEET 2013 (Karnataka)

A person holding a rifle (mass of person and rifle together is 100 kg) stands on a smooth surface and fires 10 shots horizontally, in 5 s. Each bullet has a mass of 10 g with a muzzle velocity of 800 m s^{-1}. The final velocity acquired by the person and the average force exerted on the person are :

Open solution

(A) -0.08 ms^{-1}, 16 N (B) -0.8 ms^{-1}, 8 N (C) -1.6 ms^{-1}, 16 N (D) -1.6 ms^{-1}, 8 N

Q25. NEET 2013

An explosion breaks a rock into three parts in a horizontal plane. Two of them go off at right angles to each other. The first part of mass 1 kg moves with a speed of 12 m s^{-1} and the second part of mass 2 kg moves with 8 m s^{-1} speed. If the third part flies off with 4 m s^{-1} speed, then its mass is :

Open solution

(A) 7 kg (B) 17 kg (C) 3 kg (D) 5 kg

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Q26. AIPMT 2012 Mains

Three masses are placed on the x-axis : 300 g at origin, 500 g at x = 40 cm and 400 g at x = 70cm. The distance of the centre of mass from the orgin is :

(A) 40 cm (B) 45 cm (C) 50 cm (D) 30 cm

Open solution

Q27. AIPMT 2012 Prelims

Two spheres A and B of masses m_1 and m_2 respectively collide. A is at rest initially and B is moving with velocity v along x-axis. After collision B has a velocity $\frac{v}{2}$ in a direction perpendicular to the original direction. The mass A moves after collision in the direction :

(A) same as that of B

(B) opposite to that of B

(C) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ to the x-axis

(D) $\theta = \tan^{-1}\left(-\frac{1}{2}\right)$ to the x-axis

Open solution

Q28. AIPMT 2012 Prelims

Two persons of masses 55 kg and 65 kg respectively, are at the opposite ends of a boat. The length of the boat is 3.0 m and weights 100 kg. The 55 kg man walks up to the 65 kg man and sits with him. If the boat is in still water the centre of mass of the system shifts by :

(A) 3.0 m (B) 2.3 m (C) zero (D) 0.75 m

Open solution

Q29. AIPMT 2010 Prelims

Two particles which are initially at rest, move towards each other under the action of their internal attraction. If their speeds are v and 2v at any instant, then the speed of centre of mass of the system will be :

(A) 2v (B) zero (C) 1.5 v (D) v

Open solution

Q30. AIPMT 2009

An explosion blows a rock into three parts. Two parts go off at right angles to each other. These two are, 1 kg first part moving with a velocity of 12 m s^{-1} and 2 kg second part moving with a velocity 8 m s^{-1}. If the third part flies off with a velocity of 4 m s^{-1}, its mass would be :

(A) 7 kg (B) 17 kg (C) 3 kg (D) 5 kg

Open solution

Q31. AIPMT 2009

Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$, respectively. The center of mass of this system has a position vector :

(A) $-2\hat{i} - \hat{j} + \hat{k}$ (B) $2\hat{i} - \hat{j} - 2\hat{k}$

(C) $-\hat{i} + \hat{j} + \hat{k}$ (D) $-2\hat{i} + 2\hat{k}$

Open solution

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Q32. AIPMT 2008

A particle of mass m is projected with velocity v making an angle of 45° with the horizontal. When the particle lands on the level ground the magnitude of the change in its momentum will be :

(A) $mv\sqrt{2}$ (B) zero (C) 2mv (D) mv / $\sqrt{2}$

Open solution

Q33. AIPMT 2008

A shell of mass 200 gm is ejected from a gun of mass 4 kg by an explosion that generates 1.05 kJ of energy. The initial velocity of the shell is :

(A) 40 ms^{-1} (B) 120 ms^{-1} (C) 100 ms^{-1} (D) 80 ms^{-1}

Open solution

Q34. AIPMT 2006

A 0.5 kg ball moving with a speed of 12 m/s strikes a hand wall at an angle of 30° with the wall. It is reflected with the same speed at the same angle. If the ball is in contact with the wall for 0.25 seconds, the average force acting on the wall is :

(A) 96 N (B) 48 N (C) 24 N (D) 12 N

Open solution

Q35. AIPMT 2005

A bomb of mass 30 kg at rest explodes into two pieces of masses 18 kg and 12 kg. The velocity of 18 kg mass is 6 m s^{-1}. The kinetic energy of the other mass is :

(A) 324 J (B) 486 J (C) 256 J (D) 524 J

Open solution

Q36. AIPMT 2004

Consider a system of two particles having masses m_1 and m_2. If the particle of mass m_1 is pushed towards the mass centre of particles through a distance d. by what distance would be particle of mass m_2 move so as to keep the mass centre of particles at the original position?

(A) $\frac{m_1}{m_1+m_2}$d (B) $\frac{m_1}{m_2}$d

(C) d (D) $\frac{m_2}{m_1}$d

Open solution

Q37. AIPMT 2003

A stationary particle explodes into two particles of masses m_1 and m_2 which move in opposite directions with velocities v_1 and v_2. The ratio of their kinetic energies E_1/E_2 is :

(A) m_2/m_1 (B) m_1/m_2 (C) 1 (D) m_1v_2/m_2v_1

Open solution

Q38. AIPMT 2002

If kinetic energy of a body is increased by 300% then percentage change in momentum will be :

(A) 100% (B) 150% (C) 265% (D) 73.2%

Open solution

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Q39. AIPMT 2002

A rod of length is 3 m and its mass acting per unit length is directly proportional to distance x from one of its end then its centre of gravity from that end will be at :

(A) 1.5 m (B) 2 m (C) 2.5 m (D) 3.0 m.

Open solution

Q40. AIPMT 2001

A disc is rolling, the velocity of its centre of mass is v_{cm}. Which one will be correct?

(A) the velocity of highest point is 2 v_{cm} and point of contact is zero

(B) the velocity of highest point is v_{cm} and point of contact is v_{cm}

(C) the velocity of highest point is $2v_{cm}$ and point of contact is v_{cm}

(D) the velocity of highest point is $2v_{cm}$ and point of contact is $2v_{cm}$.

Open solution

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Answer Key

1. **D**	2. **D**	3. **C**	4. **B**	5. **A**
6. **B**	7. **C**	8. **B**	9. **A**	10. **C**
11. **B**	12. **C**	13. **B**	14. **C**	15. **B**
16. **D**	17. **A**	18. **C**	19. **B**	20. **A**
21. **A**	22. **C**	23. **B**	24. **B**	25. **D**
26. **A**	27. **D**	28. **C**	29. **B**	30. **D**
31. **A**	32. **A**	33. **C**	34. **C**	35. **B**
36. **B**	37. **A**	38. **A**	39. **B**	40. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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